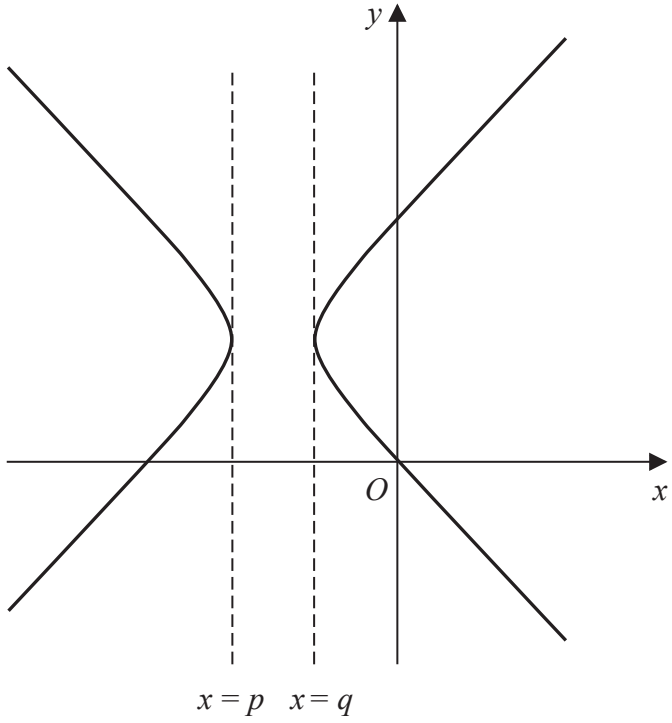



**Figure 2**

Figure 2 shows a sketch of the curve with equation

$$y^2 = 2x^2 + 15x + 10y$$

(a) Find $\frac{dy}{dx}$ in terms of $x$ and $y$.

**(4)**

The curve is not defined for values of $x$ in the interval $(p, q)$, as shown in Figure 2.

(b) Using your answer to part (a) or otherwise, find the value of $p$ and the value of $q$.

(*Solutions relying entirely on calculator technology are not acceptable*.)

**(3)**